

Lecture 9

Sequential Circuit Analysis

- ❑ Circuit Diagram to State Diagram
- ❑ State Diagram to Circuit Diagram
- ❑ Counters
- ❑ Application of Counters

CSE260 : Digital Logic Design



What is Sequential Logic ?

- ❖ Sequential circuit has additional dimension which is time
- ❖ Combinational logic only depends on current input
- ❖ Sequential circuit output depends on previous input other than current input
- ❖ More powerful than combination logic
- ❖ Able to model condition which can't be modeled by combinational logic

Circuit Diagram to State Diagram

Example 1: Draw the state diagram for the given circuit

Sequential Circuit using
D Flip Flop

Present States:

A and B

Next States:

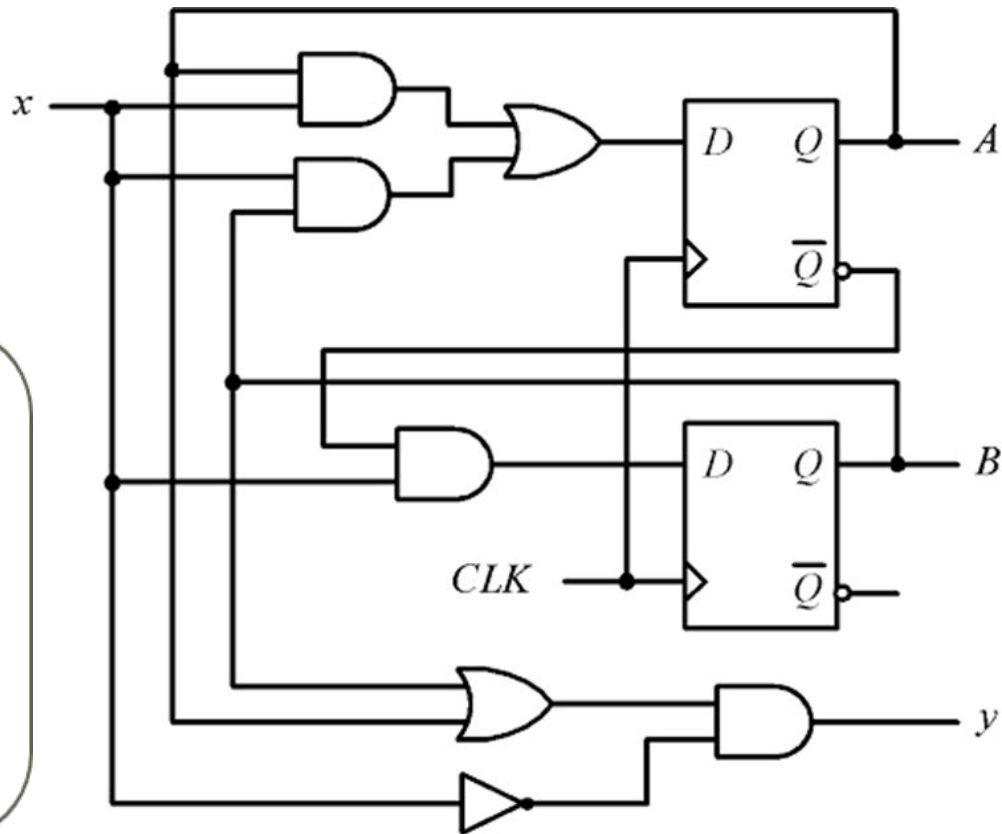
$A+$ and $B+$

Input:

DA, DB, x

Output:

y



Sequential Circuit to State Diagram Design Workflow

Steps:		Description:
1	Identify Flip-Flop Type and Equations	Extract the input equations of the Flip-Flops and external output(s) from the given circuit.
2	Build State Table	Construct the state table using the Flip-Flop Input and output equations found in step-1 .
		The state table should contain the following: Present States, external input(s), FF input and outputs.
3	Determine Next States	Compute the next states from the FF inputs and present state with the help of Truth Tables .
4	Draw State Diagram	Convert the State Table into a graphical State Diagram with labeled input transitions and output(s) if available.

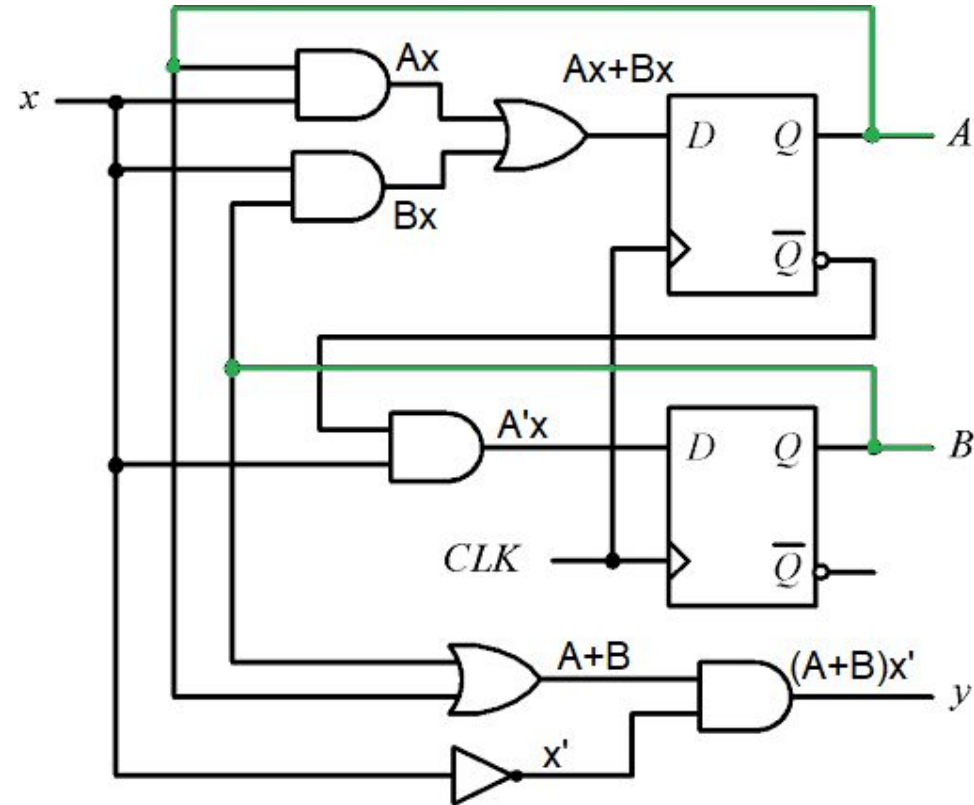
Example 1: Draw the state diagram for the given circuit

1. Find out the input equations of the flip-flop and the output equation

$$DA = Ax + Bx$$

$$DB = A'x$$

$$y = (A+B)x'$$



Example 1: Draw the state diagram for the given circuit

2. Construct the state table according to the flip-flop input and output equations

$$DA = Ax + Bx$$

$$DB = A'x$$

$$y = (A+B)x'$$

When $A=0, B=0, x=0$, When $A=0, B=0, x=1$,

$$DA = 0.0 + 0.0 = 0$$

$$DA = 0.1 + 0.1 = 0$$

$$DB = 0'.0 = 0$$

$$DB = 0'.1 = 1$$

$$y = (0+0).0' = 0$$

$$y = (0+0).1' = 0$$

A	B	x	DA	DB	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

Example 1: Draw the state diagram for the given circuit

3. Find out the next states from the flip flop input values

D Flip-Flop Truth Table

D	Q
0	0
1	1

$\therefore DA = A+$

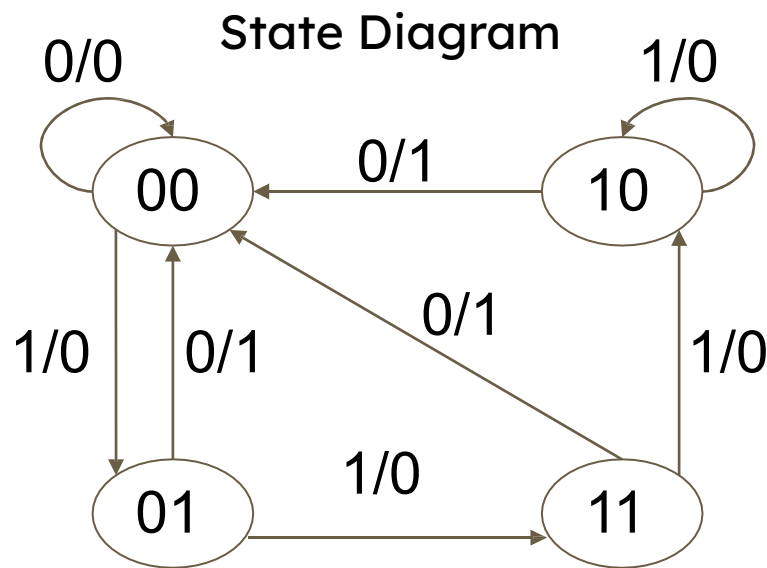
$\therefore DB = B+$

A	B	x	DA	DB	y	A+	B+
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	1	0	0	1	0

Example 1: Draw the state diagram for the given circuit

A	B	x	A+	B+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

4. Draw the state diagram from the state table



Example 2: Draw the state diagram for the given circuit

Sequential Circuit using
JK Flip Flop

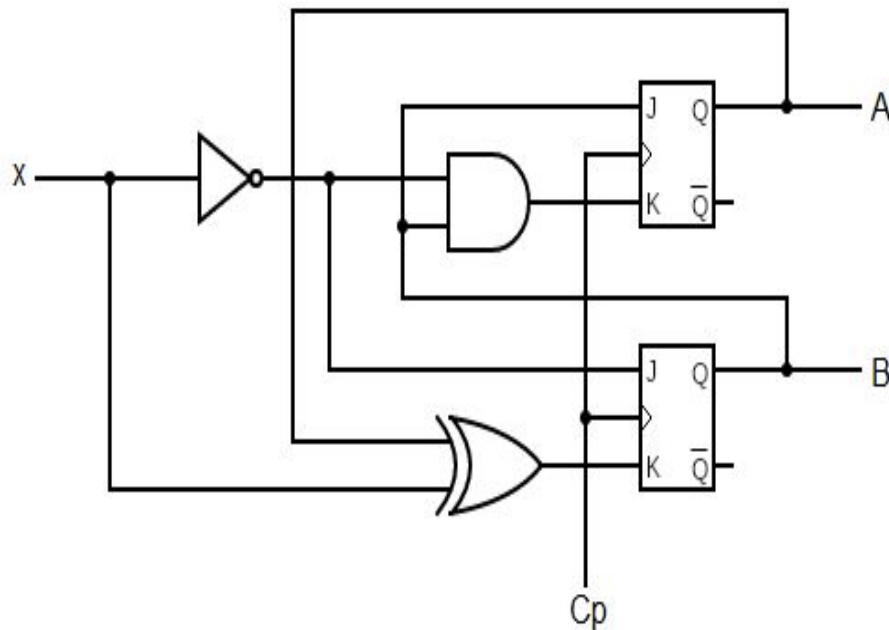
Present States: **Next States:**

A and B

$A+$ and $B+$

Input:

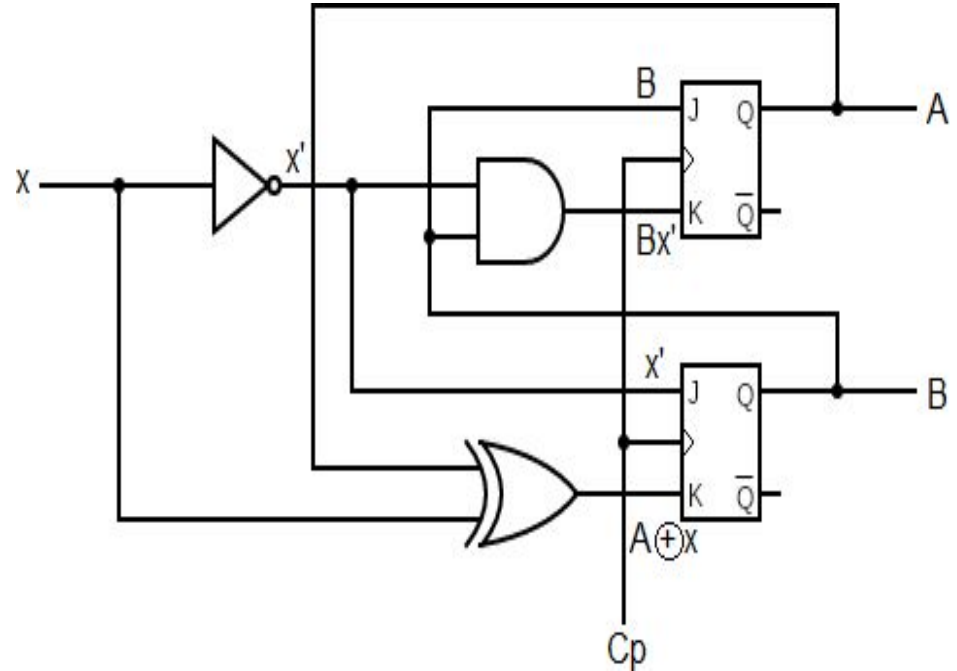
J_A, K_A, J_B, K_B, x



Example 2: Draw the state diagram for the given circuit

- Find out the input equations of the flip-flop and the output equation

$$\begin{aligned} \mathbf{JA} &= \mathbf{B} \\ \mathbf{KA} &= \mathbf{Bx'} \\ \mathbf{JB} &= \mathbf{x'} \\ \mathbf{KB} &= \mathbf{A \text{ xor } X} \end{aligned}$$



Example 2: Draw the state diagram for the given circuit

2. Construct the state table according to the flip-flop input and output equations

$$\begin{aligned} \mathbf{JA} &= \mathbf{B} \\ \mathbf{KA} &= \mathbf{Bx'} \\ \mathbf{JB} &= \mathbf{x'} \\ \mathbf{KB} &= \mathbf{A \text{ xor } X} \end{aligned}$$

When $A=0, B=0, x=0$,

$$JA = 0$$

$$KA = 0.0' = 0$$

$$JB = 0' = 1$$

$$KB = 0 \text{ xor } 0 = 0$$

When $A=0, B=0, x=1$,

$$JA = 0$$

$$KA = 0.1' = 0$$

$$JB = 1' = 0$$

$$KB = 0 \text{ xor } 0 = 0$$

A	B	x	JA	KA	JB	KB
0	0	0	0	0	1	0
0	0	1	0	0	0	1
0	1	0	1	1	1	0
0	1	1	1	0	0	1
1	0	0	0	0	1	1
1	0	1	0	0	0	0
1	1	0	1	1	1	1
1	1	1	1	0	0	0

Example 2: Draw the state diagram for the given circuit

3. Find out the next states from the flip flop input values

JK Flip-Flop Truth Table

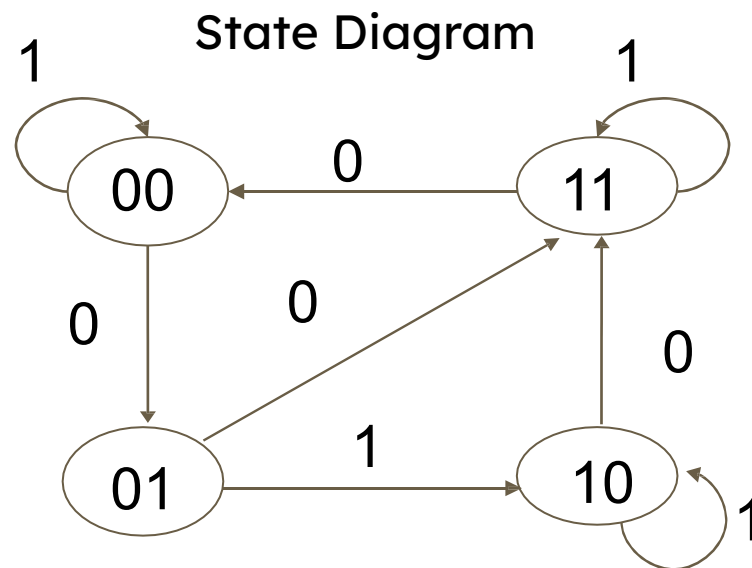
J	K	Q
0	0	Memory/ No Change
0	1	0
1	0	1
1	1	Toggle

A	B	x	JA	KA	JB	KB	A+	B+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	1	0
1	0	0	0	0	1	1	1	1
1	0	1	0	0	0	0	1	0
1	1	0	1	1	1	1	0	0
1	1	1	1	0	0	0	1	1

Example 2: Draw the state diagram for the given circuit

A	B	x	A+	B+
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

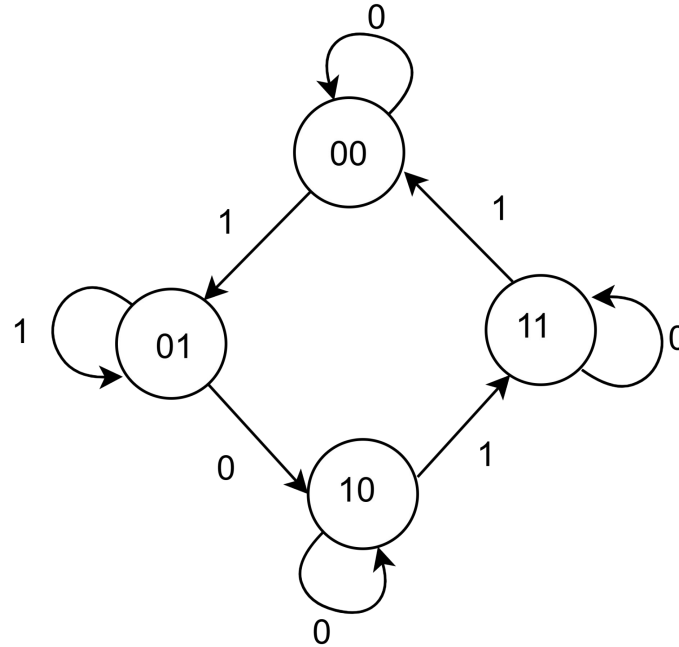
4. Draw the state diagram from the state table



State Diagram to Circuit Diagram

Design Example 1:

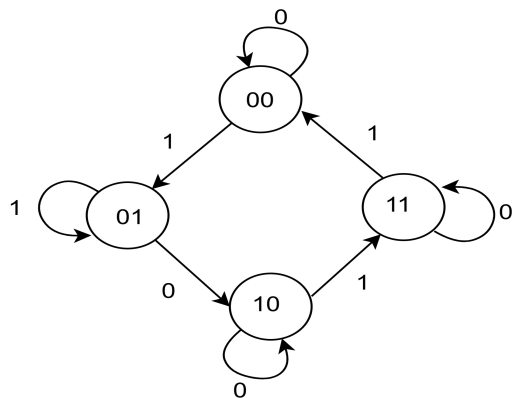
Given state diagram as follows, get the sequential circuit using JK Flip Flop



State Diagram to Sequential Circuit Design Workflow

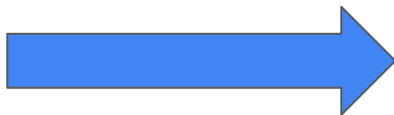
Steps:		Description:
1	Build State table	List Present and Next States Transitions with corresponding inputs. Include outputs if required
2	Determine Number of Flip-flops	If the number of states = n , then the number of flip-flops (m) required will be: $m = \lceil \log_2(n) \rceil$
3	Create Excitation Table	Determine and Populate Flip-Flop input columns based on present and next states using the excitation table .
4	Derive Input Functions using K-Map	Find each FF input and output function using n -variable K-Map. [n depends on the total number of present states and inputs]
5	Design Logic Diagram	Draw sequential circuit using logic gates and flip-flops based on the equations found in Step-4

Design Example 1: State Table



Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-Flop's
excitation table



Present State	Next State	
	$X = 0$	$X = 1$
AB	A^+B^+	A^+B^+
00	00	01
01	10	01
10	10	11
11	11	00

Present State		Input	Next State		Flip-Flop Inputs			
A	B		A^+	B^+	JA	KA	JB	KB
0	0	0	0	0	0	x	0	x
0	0	1	0	1	0	x	1	x
0	1	0	1	0	1	x	x	1
0	1	1	0	1	0	x	x	0
1	0	0	1	0	x	0	0	x
1	0	1	1	1	x	0	1	x
1	1	0	1	1	x	0	x	0
1	1	1	0	0	x	1	x	1

Design Example 1: Get FF input function using K-Map

Present State		Input	Next State		Flip-Flop Inputs			
A	B	X	A ⁺	B ⁺	JA	KA	JB	KB
0	0	0	0	0	0	x	0	x
0	0	1	0	1	0	x	1	x
0	1	0	1	0	1	x	x	1
0	1	1	0	1	0	x	x	0
1	0	0	1	0	x	0	0	x
1	0	1	1	1	x	0	1	x
1	1	0	1	1	x	0	x	0
1	1	1	0	0	x	1	x	1

		Bx			
A	0	00	01	11	10
	1				
	0		1	x	x
	1		1	x	x

$$JB = x$$

		Bx			
A	0	00	01	11	10
	1				
	0	x	x		1
	1	x	x	1	

$$KB = (A \oplus x)'$$

		Bx			
A	0	00	01	11	10
	1				
	0				1
	1	x	x	x	x

$$JA = B.x'$$

		Bx			
A	0	00	01	11	10
	1				
	0	x	x	x	x
	1			1	

$$KA = B.x$$

Design Example 1:

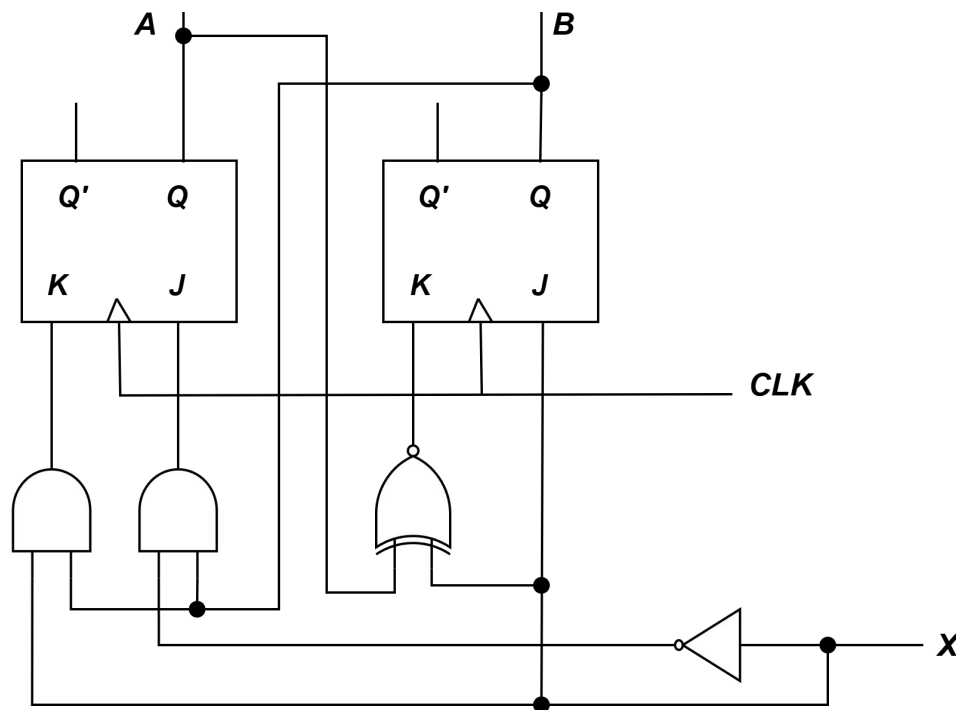
Input Flip-Flop Functions:

1. $JA = B.x'$
2. $KA = B.x$
3. $JB = x$
4. $KB = (A \oplus x)'$

Note:

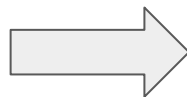
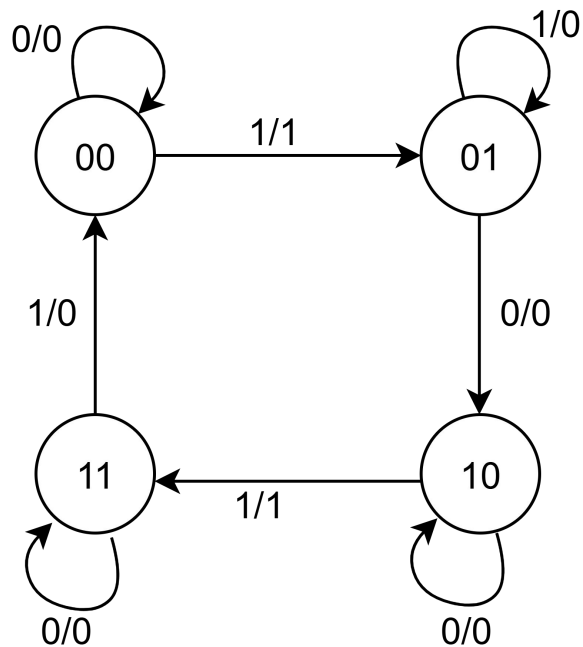
You can draw the circuit any way you wish (Horizontally or Vertically)

Logic Diagram



Design Example 2:

Design, using D flip-flop, circuit is based on state table below.
 (Exercise: How if using JK flip-flop)



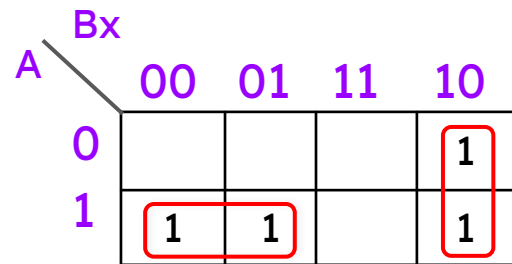
Present State		Input	Next State		Output
A	B	X	A ⁺	B ⁺	Y
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	1	0
1	1	1	0	0	0

Design Example 2: Determine FF- Input and Y output

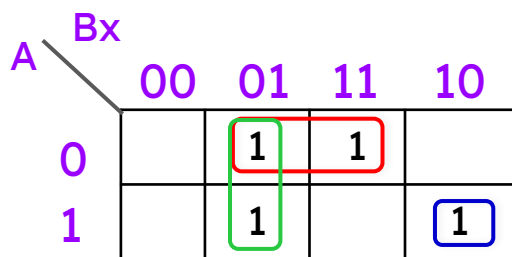
Present State		Input	Next State		Output	Inputs	
A	B	X	A ⁺	B ⁺	Y	DA	DB
0	0	0	0	0	0	0	0
0	0	1	0	1	1	0	1
0	1	0	1	0	0	1	0
0	1	1	0	1	0	0	1
1	0	0	1	0	0	1	0
1	0	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	1	1	0	0	0	0	0

Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1

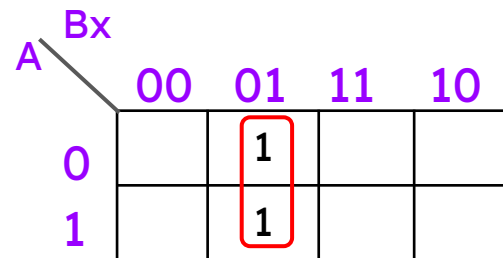
D - Flip Flop



$$DA = A.B' + B.x'$$



$$DB = A'.x + B'.x + A.B.x'$$



$$Y = B'.x$$

Design Example 2:

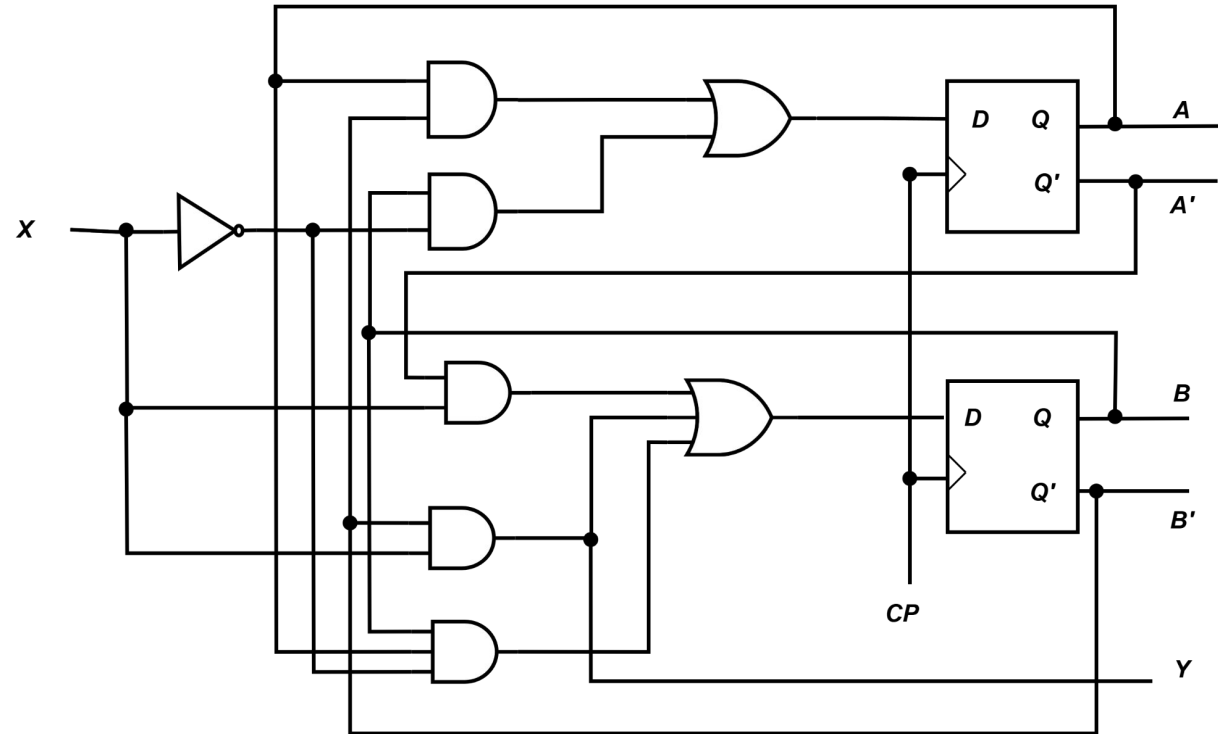
Input Flip-Flop Functions:

$$DA = A.B' + B.x'$$

$$DB = A.x + B'.x + A.B.x'$$

$$Y = B'.x$$

Logic Diagram





Introduction: Counters

- ❖ **Counters** are circuits that cycle through a specified number of states.
- ❖ Two types of counters:
 - synchronous (parallel) counters
 - asynchronous (ripple) counters
- ❖ Ripple counters allow some flip-flop outputs to be used as a source of clock for other flip-flops.
- ❖ Synchronous counters apply the same clock to all flip-flops.
- ❖ **We will cover only synchronous this time**

Synchronous (Parallel) Counters

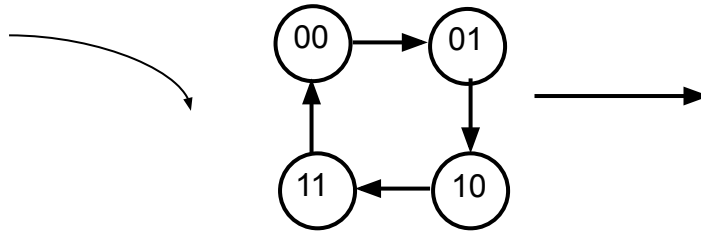
- ❖ **Synchronous (parallel) counters:** the flip-flops are clocked at the same time by a common clock pulse.
- ❖ We can design these counters using the sequential logic design process.
- ❖ Example: 2-bit synchronous binary counter (using T flip-flops, or JK flip-flops with identical J,K inputs).



Synchronous (Parallel) Counters

Question: Design a 2-bit synchronous binary counter (using T flip-flops).

Start by drawing this



Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table of T Flip-Flop

Present State		Next State		Flip-Flop Inputs	
A_1	A_0	A_1^+	A_0^+	TA_1	TA_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

Synchronous (Parallel) Counters

Question: Design a 2-bit synchronous binary counter (using T flip-flops).

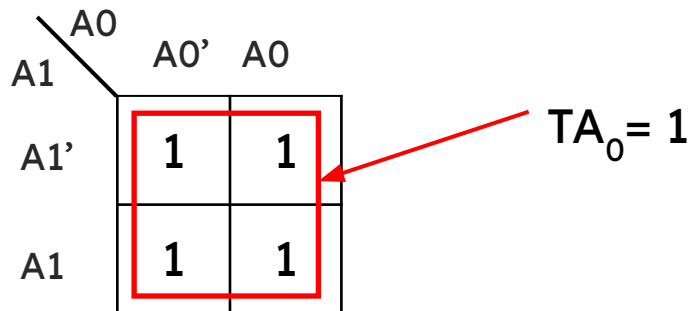
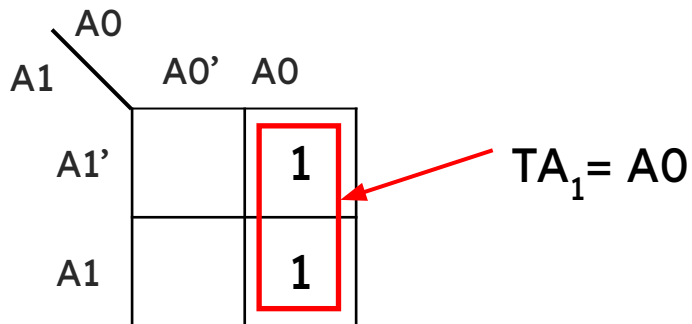
Present State		Next State		Flip-Flop Inputs	
A_1	A_0	A_1^+	A_0^+	TA_1	TA_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

Next, find out the equations for the flip-flop inputs.

$$TA_1 = A_1'A_0 + A_1A_0 = A_0 (A_1' + A_1) = A_0$$

$$TA_0 = 1$$

FYI, these can also be calculated using K-Maps

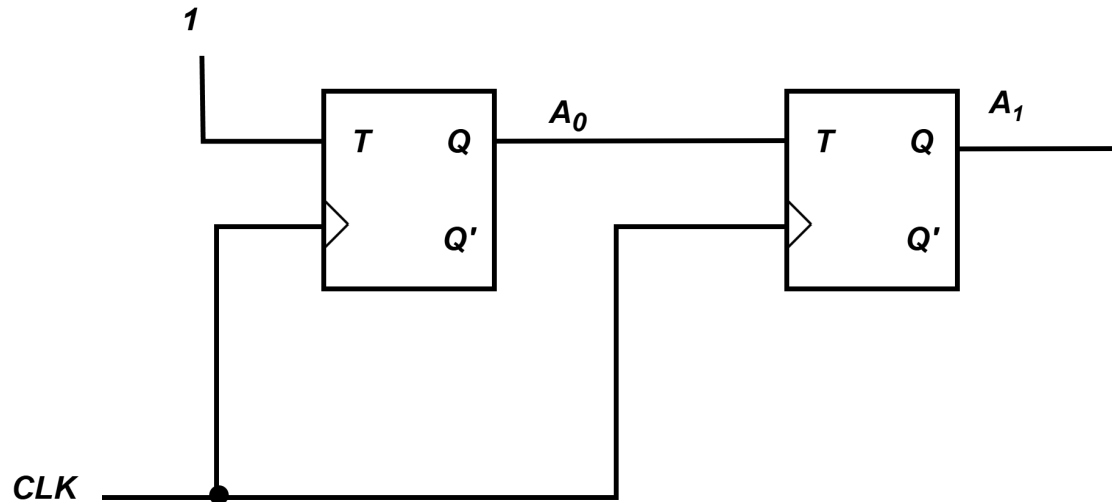


Synchronous (Parallel) Counters

$$TA_1 = A_1'A_0 + A_1A_0 = A_0 (A_1' + A_1) = A_0$$

$$TA_0 = 1$$

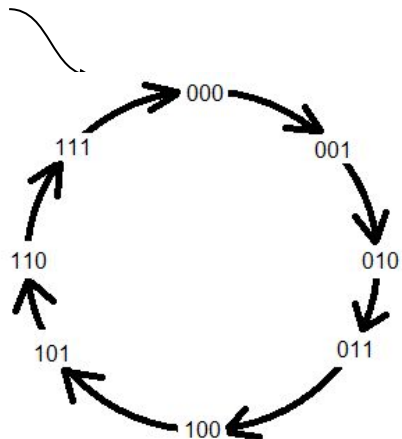
Finally, draw the circuit diagram



Synchronous (Parallel) Counters

Question: Design a 3-bit synchronous binary counter (using T FF).

Start by drawing this



Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2^+	A_1^+	A_0^+	TA_2	TA_1	TA_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table of T Flip-Flop

Synchronous (Parallel) Counters

A1A0

A2

	00	01	11	10
0			1	
1			1	

$TA_2 = A_1 \cdot A_0$

A1A0

A2

	00	01	11	10
0		1	1	
1		1	1	

$TA_1 = A_0$

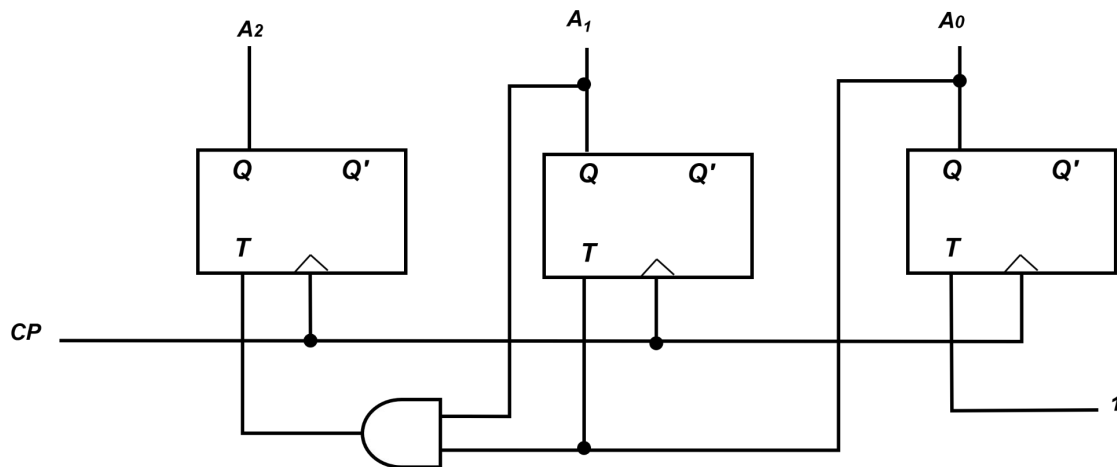
A1A0

A2

	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$TA_0 = 1$

Draw the circuit
Diagram



Equations and diagram will look like the following:

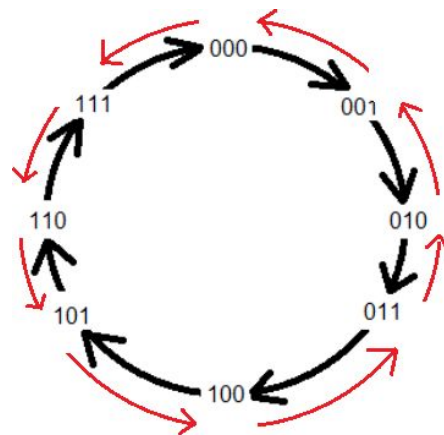
$$\tau A_0 = 1$$


Up/Down Synchronous Counters

- ❖ **Up/down synchronous counter:** a bidirectional counter that is capable of counting either up or down.
- ❖ An input (control) line Up/Down (or simply Up) specifies the direction of counting.
 - Up/Down = 0 → Count upward
 - Up/Down = 1 → Count downward

3- bit Up/Down Synchronous Counters

U/D	Present State			Next State			Flip-Flop Inputs		
M	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	TQ_2	TQ_1	TQ_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1



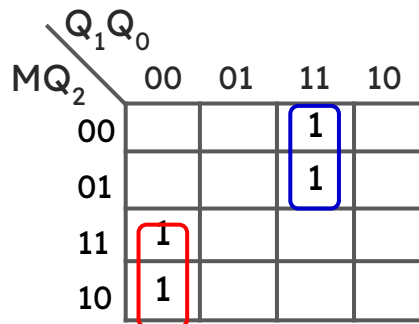
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table of
T Flip-Flop

[Reference Video](#)

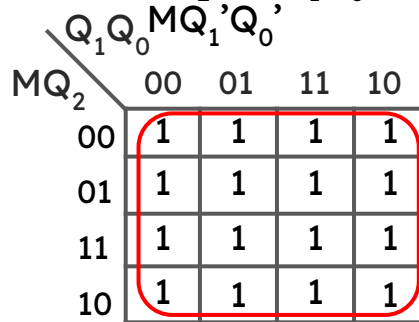
3- bit Up/Down Synchronous Counters

U/D	Present State			Next State			Flip-Flop Inputs		
M	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	TQ_2	TQ_1	TQ_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	0	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1



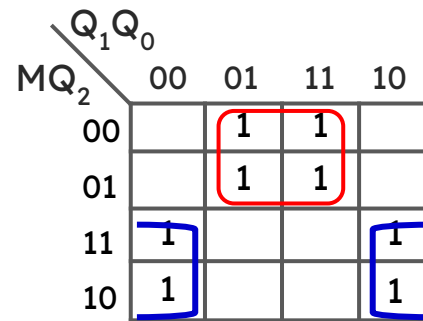
Karnaugh map for TQ_2 with variables Q_1 and Q_0 . The map shows 1s at (1,1), (1,0), (0,1), and (0,0), which are grouped by a blue rectangle.

$$TQ_2 = M'Q_1Q_0 +$$



Karnaugh map for TQ_1 with variables Q_1 and Q_0 . The map shows 1s at (1,1), (1,0), (0,1), and (0,0), which are grouped by a red rectangle.

$$TQ_0 = 1$$



Karnaugh map for TQ_0 with variables Q_1 and Q_0 . The map shows 1s at (1,1), (1,0), (0,1), and (0,0), which are grouped by a red rectangle.

$$TQ_1 = M'Q_0 + MQ_0'$$

[Reference Video](#)

(The number of present state will be equal to the number of flip-flops in the circuit)





Practice Problems using Counters

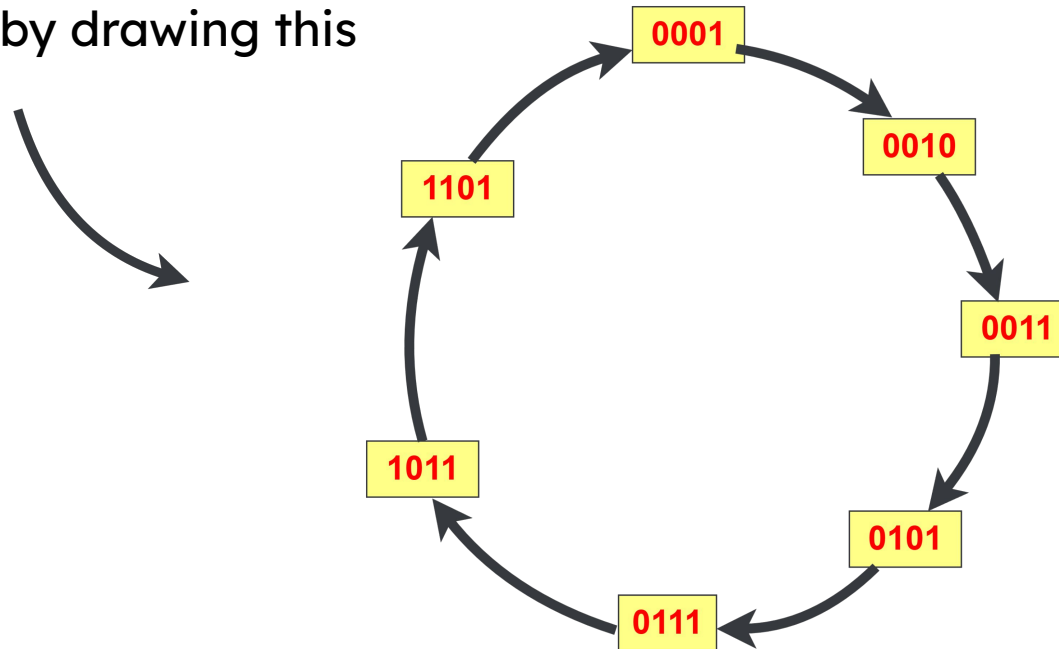


Practice Problem - 1

Implement the following counter using T FF:

1→2→3→5→7→11→13→1

Start by drawing this



Practice Problem - 1

Present State				Next State				Flip-Flop Inputs			
A_3	A_2	A_1	A_0	A_3^+	A_2^+	A_1^+	A_0^+	TA_3	TA_2	TA_1	TA_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	1	0	1	1	0
0	1	0	0	0	0	0	1	0	1	0	1
0	1	0	1	0	1	1	1	0	0	1	0
0	1	1	0	0	0	0	1	0	1	1	1
0	1	1	1	1	0	1	1	1	1	0	0
1	0	0	0	0	0	0	1	1	0	0	1
1	0	0	1	0	0	0	1	1	0	0	0
1	0	1	0	0	0	0	1	1	0	1	1
1	0	1	1	1	1	0	1	0	1	1	0
1	1	0	0	0	0	0	1	1	1	0	1
1	1	0	1	0	0	0	1	1	1	0	0
1	1	1	0	0	0	0	1	1	1	1	1
1	1	1	1	0	0	0	1	1	1	1	0

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

For present states not mentioned in the question, next state will be the **INITIAL STATE** [0001 in this question].

It's important to show the states that are not mentioned in the question. Otherwise, the circuit will stop working if any of those states show up.

Practice Problem - 1

Present State				Next State				Flip-Flop Inputs			
A_3	A_2	A_1	A_0	A_3^+	A_2^+	A_1^+	A_0^+	TA_3	TA_2	TA_1	TA_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	1	0	1	1	0
0	1	0	0	0	0	0	1	0	1	0	1
0	1	0	1	0	1	1	1	0	0	1	0
0	1	1	0	0	0	0	1	0	1	1	1
0	1	1	1	1	0	1	1	1	1	0	0
1	0	0	0	0	0	0	1	1	0	0	1
1	0	0	1	0	0	0	1	1	0	0	0
1	0	1	0	0	0	0	1	1	0	1	1
1	0	1	1	1	1	0	1	0	1	1	0
1	1	0	0	0	0	0	1	1	1	0	1
1	1	0	1	0	0	0	1	1	1	0	0
1	1	1	0	0	0	0	1	1	1	1	1
1	1	1	1	0	0	0	1	1	1	1	0

A_3A_2 \ A_1A_0
 $A_1'A_0'$ $A_1'A_0$ A_1A_0 A_1A_0'

$A_3'A_2'$				
$A_3'A_2$			1	
A_3A_2	1	1	1	1
A_3A_2'	1	1		1

$$TA_3 = A_3A_0' + A_3A_1' + A_2A_1A_0$$

A_3A_2 \ A_1A_0
 $A_1'A_0'$ $A_1'A_0$ A_1A_0 A_1A_0'

$A_3'A_2'$			1	
$A_3'A_2$	1		1	1
A_3A_2	1	1	1	1
A_3A_2'			1	

$$TA_2 = A_2A_0' + A_3A_2 + A_1A_0$$

Practice Problem - 1

Present State				Next State				Flip-Flop Inputs			
A_3	A_2	A_1	A_0	A_3^+	A_2^+	A_1^+	A_0^+	TA_3	TA_2	TA_1	TA_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	1	0	1	1	0
0	1	0	0	0	0	0	1	0	1	0	1
0	1	0	1	0	1	1	1	0	0	1	0
0	1	1	0	0	0	0	1	0	1	1	1
0	1	1	1	1	0	1	1	1	1	0	0
1	0	0	0	0	0	0	1	1	0	0	1
1	0	0	1	0	0	0	1	1	0	0	0
1	0	1	0	0	0	0	1	1	0	1	1
1	0	1	1	1	1	0	1	0	1	1	0
1	1	0	0	0	0	0	1	1	1	0	1
1	1	0	1	0	0	0	1	1	1	0	0
1	1	1	0	0	0	0	1	1	1	1	1
1	1	1	1	0	0	0	1	1	1	1	0

A_3A_2 \ A_1A_0

	$A_1'A_0'$	$A_1'A_0$	A_1A_0	A_1A_0'
$A_3'A_2'$		1	1	
$A_3'A_2$	1			1
A_3A_2			1	1
A_3A_2'			1	1

$$TA_1 = A_3'A_1'A_0 + A_3'A_2'A_0 + A_2A_1A_0' + A_3A_1$$

A_3A_2 \ A_1A_0

	$A_1'A_0'$	$A_1'A_0$	A_1A_0	A_1A_0'
$A_3'A_2'$	1	1		1
$A_3'A_2$	1			1
A_3A_2	1			1
A_3A_2'	1			1

$$TA_0 = A_0 + A_3'A_2'A_1'$$



Practice Problem - 1

$$TA_3 = A_3 A_0' + A_3 A_1' + A_2 A_1 A_0$$

$$TA_2 = A_2 A_0' + A_3 A_2 + A_1 A_0$$

$$TA_1 = A_3' A_1' A_0 + A_3' A_2' A_0 + A_2 A_1 A_0' + A_3 A_1$$

$$TA_0 = A_0 + A_3' A_2' A_1'$$

Draw the circuit yourself (Self Study)

Practice Problem - 2

Implement the following counter using D FF:
Green→Yellow→Red→Yellow→Green

Let's assume Green = 00, Yellow = 01, Red = 10

Present State			Next State		Flip-Flop Inputs	
A	B	x	A ⁺	B ⁺	DA	DB
0	0	0	0	1	0	1
0	0	1	0	1	0	1
0	1	0	0	0	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	1	0	1	0	1
1	1	0	x	x	x	x
1	1	1	x	x	x	x

Find out equations for
Flip-Flop inputs

A \ Bx				
	00	01	11	10
0			1	
1			x	x

$$DA = Bx$$

A \ Bx				
	00	01	11	10
0	1	1		
1	1	1	x	x

$$DB = B'$$

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation table of
D Flip-Flop

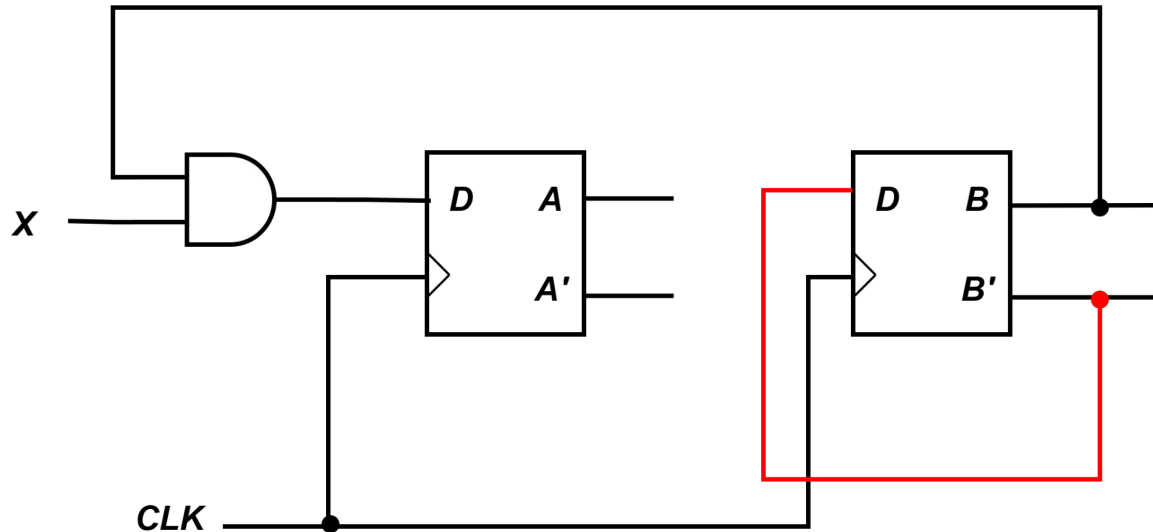
Special Note:
You may need to use more
than 1 external input
depending upon the question



Practice Problem - 2

Draw the circuit diagram

(The number of present state will be equal to the number of flip-flops in the circuit)





Practice Problem - 3 (Self Study)

Implement the following counter using D FF:

Green→Yellow→Red→Yellow→Blue→Red→Yellow→Green

Try it Yourself

Check how many external inputs you will need in this case.